

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403518
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Jäderberg; Jan et al.

Method and arrangement for producing a hardened sheet metal product

Abstract

The present disclosure relates to a method and a corresponding arrangement for producing a hardened sheet metal product. The method includes placing a sheet metal piece in a heating station and heating selected areas of the sheet metal piece in the heating station by means of induction. In this process, a coil induces currents that flow in a front metal layer on a front side of the coil. Opposing ends of the front metal layer are interconnected by a low resistance short-circuiting arrangement running on a rear side of the coil. The short-circuiting arrangement comprises a material with lower resistivity than the front metal layer. The heated piece is moved to a pressing station, where it is pressed while the heated areas are cooled.

Inventors:	Jäderberg; Jan (Spånga, SE), Brandt; Jörgen (Åhus, SE)
Applicant:	TCTECH SWEDEN AB (Spånga, SE)
Family ID:	62706457
Assignee:	TC TECH SWEDEN AB (N/A, SE)
Appl. No.:	18/606020
Filed:	March 15, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240261842 A1	Aug. 08, 2024

Foreign Application Priority Data

SE	1750017-4	Jan. 11, 2017
----	-----------	---------------

Related U.S. Application Data

Publication Classification

Int. Cl.: **B21D22/02** (20060101); **C21D1/42** (20060101); **C21D1/673** (20060101); **H05B6/36** (20060101); **H05B6/40** (20060101)

U.S. Cl.:

CPC **B21D22/022** (20130101); **C21D1/42** (20130101); **C21D1/673** (20130101); **H05B6/365** (20130101); **H05B6/40** (20130101); C21D2221/10 (20130101)

Field of Classification Search

CPC: B21D (22/022); C21D (1/42); C21D (1/673); H05B (6/365); H05B (6/40)

USPC: 219/645

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4321097	12/1981	Kerr	N/A	N/A
7981350	12/2010	Jaderberg	425/174.8 E	B29C 33/04
9482304	12/2015	Battlogg et al.	N/A	N/A
2010/0326984	12/2009	Mochinaga et al.	N/A	N/A
2013/0312478	12/2012	Kobayashi	N/A	N/A
2013/0333190	12/2012	Mizumura et al.	N/A	N/A
2014/0124104	12/2013	Trippe	72/342.7	C22C 38/44
2015/0053670	12/2014	Horn et al.	N/A	N/A
2015/0107325	12/2014	Glueck et al.	N/A	N/A
2015/0321388	12/2014	Jderberg et al.	N/A	N/A
2016/0108489	12/2015	Sikora et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2137199	12/1998	AU	N/A
102921727	12/2012	CN	N/A
103447366	12/2012	CN	N/A
102012021031	12/2012	DE	C21D 1/673
2955239	12/2014	EP	N/A
2960035	12/2014	EP	N/A
S5311110	12/1977	JP	N/A
2006212690	12/2005	JP	N/A
2006212705	12/2005	JP	N/A
2006348374	12/2005	JP	N/A
2009022995	12/2008	JP	N/A
2013244510	12/2012	JP	N/A

2015039719	12/2014	JP	N/A
2016153237	12/2015	JP	N/A
2016524043	12/2015	JP	N/A
2012118223	12/2011	WO	N/A
2013002703	12/2012	WO	N/A
2014000900	12/2013	WO	N/A
2014009000	12/2013	WO	N/A
2014104963	12/2013	WO	N/A
2015197415	12/2015	WO	N/A

OTHER PUBLICATIONS

Machine Translation of DE102012021031 (Year: 2024). cited by examiner

Primary Examiner: Crabb; Steven W

Assistant Examiner: Ward; Thomas J

Attorney, Agent or Firm: Condo Roccia Koptiw LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/476,933, filed Jul. 10, 2019, which is a National Stage Entry under 35 U.S.C. § 371 of Patent Cooperation Treaty Application No. PCT/SE2018/050001, filed 2 Jan. 2018, which claims priority from Swedish Application No. 1750017-4, filed 11 Jan. 2017, the contents of each of which are hereby incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

(1) The present disclosure relates to a method for producing a hardened sheet metal product. The disclosure further relates to an arrangement and devices configured to carry out the method.

TECHNICAL BACKGROUND

(2) Hardening of sheet metal products is carried out to alter metallurgic properties of a piece of sheet metal. Traditionally, hardening has been done by heating a piece of metal in a forge and for instance cooling it quickly using water. It has recently been suggested to partially heat a piece of sheet metal using a high-power laser. One problem with such processes is that they are rather slow, as the laser beam has to sweep the entire trace that need be heated. Of course, this could be remedied to some extent by using multiple lasers, but to a much higher cost. Three to five paragraphs should do it, but this is not a hard and fast rule. May depend on the number of independent claims.

SUMMARY OF THE INVENTION

(3) One object of the present disclosure is therefore to provide a hardening method which is more efficient. This object is achieved by means of a method as defined in claim 1. More specifically, there is provided a method for producing a hardened sheet metal product comprising placing a sheet metal piece in a heating station, and heating selected areas of the sheet metal piece in the heating station by means of induction. In the heating station, a coil induces currents that flow in a front metal layer on a front side of the coil and first and second opposing ends of the front metal layer are inter-connected by a short-circuiting arrangement running on a rear side of the coil, the short-circuiting arrangement comprising a material with lower resistivity than the front metal layer. The heated sheet metal piece is moved to a pressing station, and the sheet metal piece is pressed

while cooling said heated areas.

(4) With this method, all areas of the sheet metal piece that are to be heated can be heated simultaneously which provides for a much more efficient process. Further, the heated areas may be cooled more quickly and in a much more uniform manner, which provides improved hardening properties.

(5) In one example, suitable e.g. for hardening steel, the front metal layer may comprise the sheet metal piece, i.e. the currents developing heat are induced in the piece to be processed itself.

(6) In an alternative example, the front metal layer may comprise a heating layer located directly beneath the sheet metal piece, the sheet metal piece being separable from the heating layer. This is useful for treating sheet metal with very low resistivity, such as aluminum.

(7) Inlays may be located directly beneath the front metal layer, which by conveying magnetic flows or electric currents partially reduce the development of heat in the front metal layer. This allows a pattern with any desired shape to be heated on the sheet metal piece.

(8) A corresponding production arrangement and a corresponding heating station may also be considered.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates schematically an arrangement for processing a sheet metal material according to the present disclosure.

(2) FIG. 2 illustrates the operating of a heating station of FIG. 1.

(3) FIG. 3 shows a processed piece of sheet metal.

(4) FIG. 4 illustrates schematically a stack of layers in a first example of a heating station.

(5) FIG. 5 illustrates schematically a stack of layers in a second example of a heating station.

(6) FIG. 6 illustrates varying heating over a sheet metal surface by means of inlays beneath the sheet metal piece.

DETAILED DESCRIPTION

(7) FIG. 1 illustrates schematically an arrangement **1** for processing a sheet metal material according to the present disclosure. The arrangement **1** includes a heating station **3** and a pressing station **7** or press as well as a transport device **5**, devised to rapidly transport a heated piece of sheet metal from the heating station **3** to the pressing station **7**. This may be done such that pressing can take place within five seconds from ending heating.

(8) The arrangement **1** realizes a method for producing a hardened sheet metal product that includes the following steps. First, a piece of sheet metal **9** is placed in the heating station **3** as shown in FIG. 1. This may be done manually or automatically by means of an additional transport device (not shown). As illustrated, the piece of sheet metal **9** may be flat, although this is not necessary.

(9) The heating station **3** heats the sheet metal piece, either the whole surface thereof or as illustrated in FIG. 2, a pattern **11** on that surface. The amount of heating needed depends on the metallurgic properties of the material of the piece **9**, and the desired outcome of the hardening process. For instance, if a steel is hardened and a trace with a different crystalline structure is desired along the pattern **11**, the material in the pattern could be heated to a temperature e.g. above 910° C. and then relatively quickly cooled.

(10) With reference again to FIG. 1, once the heating has taken place, in a manner described later, the piece **9** is quickly moved to the press or pressing station **7**, where the piece is pressed into a shape deviating from the previous shape, and optionally punched to provide cut-outs in the piece, as desired. Not only does this process reshape the piece **9**, but also cools the previously heated parts of the piece to provide a hardening effect, if needed. The result is a piece as shown in FIG. 3. The

hardened pattern **11** may help to provide the finished piece **13** with desired properties, for instance it may deform in a desired pattern thanks to a stiffer but more brittle material in the generated pattern **11**. This may be useful for instance for automotive products where driver and passenger safety requires that a colliding car is deformed in a predetermined way.

(11) Returning again to FIG. **1**, the transport device **5** is symbolically illustrated as a roller.

However, various devices can be used to transport the piece **9** from the heating station **3** to the pressing station **7**, such as industrial robots, etc. In some hardening processes, it may be required that the time before the heating ends and the pressing takes place is short, typically under 5 seconds which sets a requirement for the transport device **5**.

(12) The heating station **3** in FIG. **1** is illustrated as open upwards. However, it is possible, similar to in the pressing station **7** as illustrated, to provide an upper half also in the heating station **3** that presses against the lower half.

(13) The heating of the selected areas of the sheet metal piece **9** in the heating station **3** is carried out by means of induction as will now be described with reference to a first example illustrated in FIG. **4**, showing a schematic cross section through a heating station **3**.

(14) Specifically, there is used a coil **17** which is fed by a high frequency (typically in the range 20-50 kHz) alternating current pulse. The coil is made from a low resistivity material, such as aluminum or copper, and is wound around a coil carrier **15**.

(15) The coil carrier **15** may comprise a material with high resistivity, and that has a high relative magnetic permeability. Soft magnetic composites, such as for instance SOMALOY, comprising ferromagnetic granules that are sintered to a desired shape with an insulating plastic material is one example of materials suitable for this purpose.

(16) The coil **17** and the coil carrier **15** will induce strong electric currents that flow in conductive neighboring elements. A main current loop is formed by the sheet metal piece **9** and a short-circuit arrangement **23**. The short-circuit arrangement **23** interconnects opposing edges **33**, **35** of the sheet metal piece **9**, which in this example forms a front metal layer. 'Front' here relates to the surface on a front side of the coil **17** and the surface of the heating station **3** where heating is intended to take place. Opposing edges **33**, **35** of the front metal layer, where the coil turns, are interconnected by a short-circuiting arrangement **23** running on a rear side of the coil **17**. Thus, strong alternating currents will run in the direction indicated by arrows in the drawing, while likewise alternating magnetic fields run perpendicular to the currents. The currents will develop heat in this loop. However, if the sheet metal piece **9** to be heated and making up the front metal layer is a steel that allows hardening, and the short-circuiting arrangement **23** is made of e.g. copper or aluminum, most of that heat will be developed in the higher resistivity sheet metal piece **9** that becomes heated in a very efficient way. Generally, the short-circuiting arrangement **23** may comprise a material with lower resistivity than the front metal layer.

(17) An intermediate conductive layer **19** may be placed in between the coil carrier **15** and the front metal layer **9**. This intermediate layer **19** may be electrically/galvanically insulated from neighboring layers but may itself be highly conductive, for instance made of copper or aluminum and may be up to a few centimeters thick. The coil **17** induces currents in the lower face of the intermediate layer **19**, and those currents run, due to the skin effect, close to the surface of the layer **19**, along the lower face, a first end face, the upper face and a second end face back to the lower face to form a closed loop close to the outer boundaries of the intermediate layer **19**. Therefore, strong currents will be present in the top surface of the intermediate layer **19** that assist in driving currents through the front metal layer **9** by induction.

(18) A thermally insulating layer **21** may be placed beneath, typically directly beneath the front metal layer **9**. This layer serves to reduce the conduction of heat from the front metal layer **9** such that the latter can reach higher temperatures. Materials such as glass, ceramic compositions as for instance including yttrium stabilized zirconium, YSZ, or different plastic materials such as KAPTON, may be considered for this purpose.

(19) FIG. 5 illustrates schematically a stack of layers in a second example of a heating station 3'. This stack is intended for the heating of sheet metal pieces 9 which themselves have a very low resistivity, such as aluminum. In this example, a stationary front metal layer 25 is provided that is connected to the short-circuiting arrangement 23. That connection may be more or less permanent, and the stationary layer 25 may have similar properties as the sheet metal piece 9 making up the front metal layer in the example in FIG. 4, i.e. high a conductivity, but still lower than the one of the short-circuiting arrangement.

(20) In this example, therefore, it is the stationary layer 25 that is heated, and this heat in turn is conveyed to the sheet metal piece 9 stacked on the stationary layer 25. Then, the sheet metal piece 9 is separated from the stationary layer and moved to the pressing station. This arrangement for instance allows hardening of aluminum. While the heating will not be as effective as in the previous example, temperature requirements may not be as high.

(21) FIG. 6 illustrates varying of the heating over a sheet metal piece's 9 surface. This may be accomplished by means of inlays that are located directly beneath the front metal layer, whether this is the sheet metal piece 9 itself as in FIG. 4 or a stationary metal layer as in FIG. 5. The inlays may therefore be located in milled recesses in the underlying, thermally insulating layer 21. The inlays may comprise electrically highly conductive pieces 27, such as copper or aluminum. Alternatively, highly magnetically conductive pieces 29 with high relative magnetic permeability such as used in the coil carrier can be considered. The electrically conductive pieces 27 will locally divert the electric current from the front metal layer. Similarly, the highly magnetic pieces will divert the magnetic field from the front metal layer. In either case areas 31 with reduced heat development will result in the sheet metal piece, which allows selective hardening of the piece 9. Additional cooling may optionally be provided, that cools areas not intended to be heated to some extent, e.g. by means of a fluid flow, for instance in the intermediate layer 19.

(22) The present disclosure is not restricted to the described examples, and may be varied and altered in different ways within the scope of the appended claims. For instance, even if above the heating of a flat sheet metal piece is described, it could be curved or in principle have an arbitrary shape. It would also be possible to combine the heating station with the press, thereby eliminating the need of a transport device in between the heating and the pressing operations. Metal working may further be divided into a plurality of successive steps, and a heating station as disclosed above may be used also after an initial pressing operation.

Claims

1. A method for producing a hardened sheet metal product, the method comprising: placing a sheet metal piece in a heating station; heating selected areas of the sheet metal piece in the heating station by means of induction; wherein a coil induces currents that flow in a front metal layer on a front side of the coil, wherein first and second opposing ends of the front metal layer are interconnected by a short-circuiting arrangement running on a rear side of the coil, the short-circuiting arrangement comprising a material with lower resistivity than the front metal layer, wherein a thermally insulating layer is placed beneath the front metal layer, and wherein inlays are located in the thermally insulating layer beneath the front metal layer, locally diverting the electric current from the front metal layer which by conveying magnetic flows or electric currents reduce the development of heat in the front metal layer in a pattern thereon; moving the heated sheet metal piece to a pressing station; and pressing the sheet metal piece while cooling said heated areas.
2. The method according to claim 1, wherein the front metal layer comprises the sheet metal piece.
3. The method according to claim 1, wherein the front metal layer comprises a stationary heating layer located directly beneath the sheet metal piece, the sheet metal piece being separable from the stationary heating layer.
4. The method according to claim 1, wherein said inlays comprise an electrically highly conductive

material, thereby locally diverting the electric current from the front metal layer.

5. The method according to claim 1, wherein said inlays comprise a material with low relative magnetic reluctance, thereby locally diverting the magnetic field from the front metal layer.

6. An arrangement for producing a hardened sheet metal product, the arrangement comprising: a heating station configured to receive a sheet metal piece, the heating station being configured to heat selected areas of the sheet metal piece in the heating station by means of induction, and the heating station comprising a coil inducing alternating currents that flow in a front metal layer on a front side of the coil, wherein first and second opposing ends of the front metal layer are interconnected by a short-circuiting arrangement running on a rear side of the coil, the short-circuiting arrangement comprising a material with lower resistivity than the front metal layer, a thermally insulating layer placed beneath the front metal layer, and inlays located in the thermally insulating layer beneath the front metal layer, configured to locally divert the electric current from the front metal layer which by conveying magnetic flows or electric currents reduce the development of heat in the front metal layer in a pattern thereon; a pressing station configured to press the sheet metal piece while cooling the heated areas; and a transport device configured to move the heated sheet metal piece from the heating station to the pressing station.

7. The arrangement according to claim 6, wherein the front metal layer is made up by the sheet metal piece to be heated.

8. The arrangement according to claim 6, wherein the front metal layer comprises a stationary heating layer located directly beneath the sheet metal piece, the sheet metal piece being separable from the stationary heating layer.

9. The arrangement according to claim 6, wherein said inlays comprise an electrically highly conductive material configured to locally divert the electric current from the front metal layer.

10. The arrangement according to claim 6, wherein said inlays comprise a material with low relative magnetic reluctance, configured to locally diverting the magnetic field from the front metal layer.

11. A heating station for a sheet metal hardening arrangement, the heating station configured to receive a sheet metal piece and to heat selected areas of the sheet metal piece by means of induction, wherein the heating station comprises: a coil that induces alternating currents that flow in a front metal layer on a front side of the coil, wherein first and second opposing ends of the front metal layer are interconnected by a short-circuiting arrangement running on a rear side of the coil, the short-circuiting arrangement comprising a material with lower resistivity than the front metal layer; a thermally insulating layer placed beneath the front metal layer; and inlays located in the thermally insulating layer beneath the front metal layer, said inlays configured to locally divert the electric current from the front metal layer which by conveying magnetic flows or electric currents reduce the development of heat in the front metal layer in a pattern thereon.
